

# Climate 5000 Series - Gen 4

## 18K BTU Single Zone Max Performance System

### 4-Way Cassette IDU

Engineering  
Submittal  
Sheet



# BOSCH

#### Overview and Project Information



|                    |  |
|--------------------|--|
| Project name       |  |
| Location           |  |
| Architect          |  |
| Engineer           |  |
| General contractor |  |
| Reference #        |  |

#### 18K Btu Single Zone Max Performance System - 4-Way Cassette IDU

|                      |                      |
|----------------------|----------------------|
| Indoor part number   | 8-733-962-718        |
| Indoor model number  | BMS500-AAU018-1AHCXD |
| Outdoor part number  | 8-733-962-690        |
| Outdoor model number | BMS500-AAS018-1CSXHD |

#### Efficiency Data AHRI 210/240 2023

|                 |      |
|-----------------|------|
| SEER2           | 20.0 |
| EER2            | 12.5 |
| HSPF2 Region IV | 12.3 |
| HSPF2 Region V  | 9.5  |
| COP(47F)        | 3.18 |
| COP (5F)        | 1.90 |

#### Performance Data AHRI 210/240 2023

|         |                      |            |
|---------|----------------------|------------|
| Cooling | Capacity Range       | 7000~18500 |
|         | Rated Capacity       | 16000      |
|         | Current              | 8.0        |
|         | Input Power          | 1280       |
| Heating | Capacity Range       | 8700~20000 |
|         | Rated Capacity (47F) | 19000      |
|         | Rated Capacity (5F)  | 17000      |
|         | Current              | 11.0       |
|         | Input Power          | 1750       |

#### Electrical Data

|                                          |                  |
|------------------------------------------|------------------|
| Power supply (V / Ø / Hz)                | 208-230 / 1 / 60 |
| Min. circuit ampacity indoor/outdoor (A) | 3.0 / 19.0       |
| Max. fuse indoor/outdoor (A)             | 15.0 / 20.0      |

#### Compressor Data

|                                  |             |
|----------------------------------|-------------|
| Type:                            | Rotary      |
| Refrigerant oil /oil charge (ml) | VG74 / 620  |
| Input (W)                        | 805 / 2045  |
| Rated current (RLA/A)            | 4.15 / 9.30 |

#### Piping

|                                                |              |
|------------------------------------------------|--------------|
| Refrigerant precharge (oz/kg)                  | 55.38 / 1.57 |
| Length for precharge (ft/m)                    | 24.6 / 7.5   |
| Liquid side (in/mm)                            | 1/4" / Ø6.35 |
| Gas side (in/mm)                               | 1/2" / Ø12.7 |
| Additional Charge of Refrigerant (oz/ft)/(g/m) | 0.16 / 15    |
| Max. refrigerant pipe length (ft/m)            | 98.40 / 30   |
| Max. difference in height (ft/m)               | 65.60 / 20   |

#### Operation Range

|     |                   |                  |
|-----|-------------------|------------------|
| ODU | Cooling (°F / °C) | -22~122 / -30~50 |
|     | Heating (°F / °C) | -22~75 / -30~24  |
| IDU | Cooling (°F / °C) | 60~90 / 16~32    |
|     | Heating (°F / °C) | 32~86 / 0~30     |

#### System Data

|                           |                                             |                                       |
|---------------------------|---------------------------------------------|---------------------------------------|
| Refrigerant type          |                                             | R454B                                 |
| Design pressure (PSIG)    |                                             | 550 / 340                             |
| Dehumidification (pts/hr) |                                             | 4.23                                  |
| IDU                       | Fan motor model                             | ZKFN-45-8-2                           |
|                           | Fan motor qty (ea)                          | 1                                     |
|                           | Fan motor input (W)                         | 50.0                                  |
|                           | Fan speed (RPM)                             | 800 / 750 / 700                       |
|                           | Air flow, Turbo / H / M / L / Si (CFM)      | 447.3 / 388.5 / 317.8 / 194.2 / 147.2 |
|                           | Approx. Air Throw Range ft (m)              | 9.8 / 3                               |
|                           | Noise level, Turbo / H / M / L / Si (dB(A)) | 48 / 44.5 / 41.0 / 27.5 / 23.5        |
|                           | Net weight (lbs / kg)                       | 35.27 / 16.0                          |
|                           | Gross weight (lbs / kg)                     | 41.89 / 19.0                          |
|                           | Panel net weight (lbs / kg)                 | 5.95 / 2.7                            |
| ODU                       | Gross panel weight (lbs / kg)               | 9.48 / 4.3                            |
|                           | Drain pipe diameter (mm)                    | ODØ25mm                               |
|                           | Fan motor model                             | ZKFN-80-8-3                           |
|                           | Fan motor qty (ea)                          | 1                                     |
|                           | Fan motor input (W)                         | 115                                   |
|                           | Fan motor speed (RPM)                       | 780 / 700 / 550                       |
|                           | Air flow, Max (CFM)                         | 1765.8                                |
|                           | Noise level, Max (dB(A))                    | 57.5                                  |
|                           | Net weight (lbs / kg)                       | 99.87 / 45.3                          |
|                           | Gross weight (lbs / kg)                     | 107.8 / 48.9                          |

#### Accessories

|                          |                              |
|--------------------------|------------------------------|
| BMS-CCG-XXD (8733962708) | Decorative Grille - 9K ~ 18K |
| BMS-WT2-XXD (8733962707) | Wired Wall Thermostat        |
| BMS-DRY-XXD (8733969263) | 24V Interface                |
| BMS-PK2-7X1 (8733951011) | 25ft Lineset                 |

#### Standard Parts

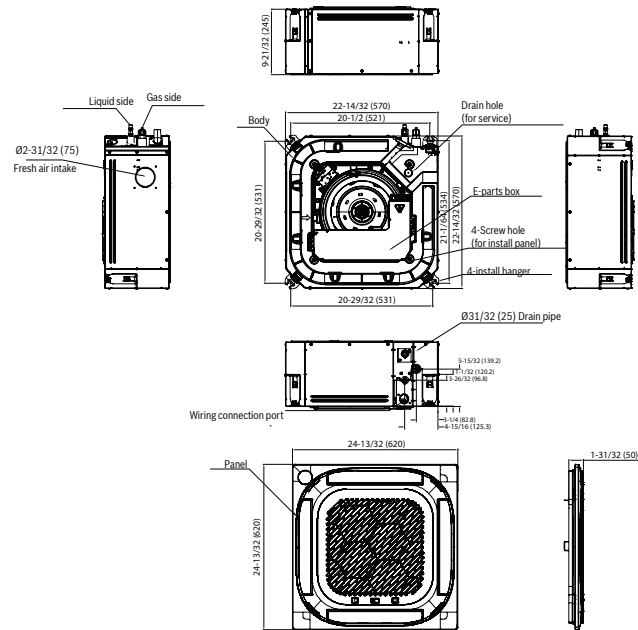
|               |                |
|---------------|----------------|
| 8-733-969-065 | Remote Control |
|---------------|----------------|

# Climate 5000 Series - Gen 4

18K Btu Single Zone Max Performance System - 4-Way Cassette IDU

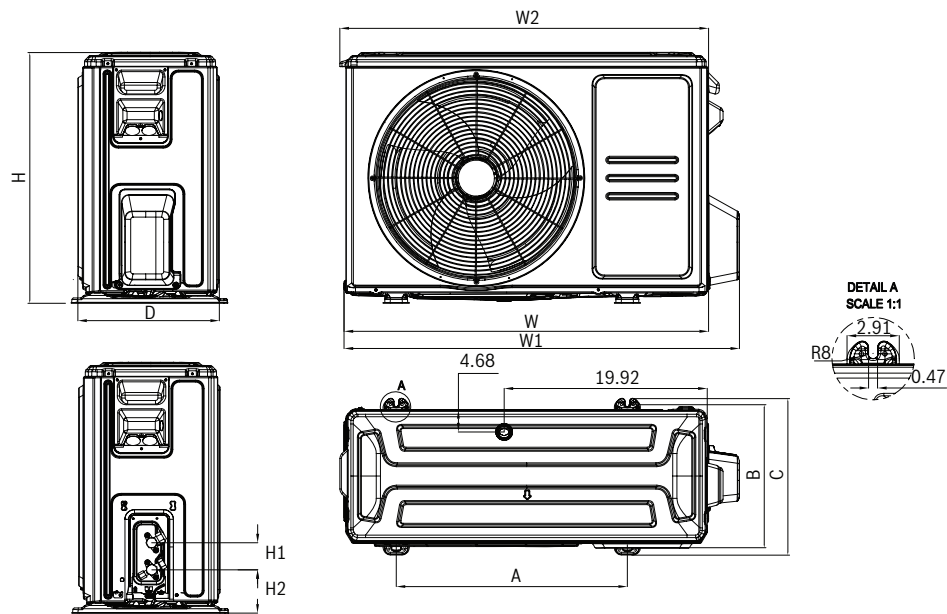


## Indoor Unit Dimensions and Connections



| Model                | W (in) | D (in) | H (in) |
|----------------------|--------|--------|--------|
| BMS500-AAU018-1AHCXD | 22.44  | 22.44  | 9.65   |

## Outdoor Unit Dimensions and Connections



| Model                | W (in) | D (in) | H (in) | W1 (in) | W2 (in) | A (in) | B (in) | C (in) | H1 (in) | H2 (in) |
|----------------------|--------|--------|--------|---------|---------|--------|--------|--------|---------|---------|
| BMS500-AAS018-1CSXHD | 35.04  | 13.46  | 26.50  | 37.6    | 35.24   | 26.1   | 13.7   | 14.96  | 2.36    | 4.25    |

## Climate 5000 Series - Gen 4

## 18K Btu Single Zone Max Performance System - 4-Way Cassette IDU


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| Cooling Performance - Indoor Dry Bulb Temperature at 447 CFM |             |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |
|--------------------------------------------------------------|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| ODU DB(°F)                                                   | IDU WB (°F) | 60.8°F(16.0°C)  |                 |                 |                 |                 | 64.4°F(18.0°C)  |                 |                 |                 |                 | 67°F(19.4°C)    |                 |                 |                 |                 | 71.6°F(22.0°C)  |                 |                 |                 |                 |
|                                                              | IDU DB (°F) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) |
| -22°F (-30.0°C)                                              | TC          | 13992           | 14011           | 14026           | 14027           | 14046           | 14836           | 14853           | 14868           | 14871           | 14889           | 15722           | 15742           | 15760           | 15763           | 15781           | 16669           | 16687           | 16706           | 16708           | 16729           |
|                                                              | S/T         | 0.60            | 0.71            | 0.80            | 0.80            | 0.91            | 0.57            | 0.67            | 0.76            | 0.76            | 0.86            | 0.53            | 0.63            | 0.71            | 0.71            | 0.81            | 0.50            | 0.59            | 0.67            | 0.67            | 0.77            |
|                                                              | PI          | 0.54            | 0.54            | 0.54            | 0.54            | 0.54            | 0.57            | 0.58            | 0.58            | 0.58            | 0.58            | 0.61            | 0.61            | 0.61            | 0.61            | 0.61            | 0.65            | 0.65            | 0.65            | 0.65            | 0.65            |
| -13°F (-25.0°C)                                              | TC          | 15158           | 15179           | 15195           | 15196           | 15216           | 16072           | 16091           | 16107           | 16110           | 16129           | 17032           | 17054           | 17073           | 17077           | 17096           | 18058           | 18078           | 18098           | 18101           | 18123           |
|                                                              | S/T         | 0.61            | 0.72            | 0.82            | 0.82            | 0.93            | 0.58            | 0.68            | 0.77            | 0.77            | 0.88            | 0.54            | 0.64            | 0.73            | 0.73            | 0.83            | 0.51            | 0.60            | 0.68            | 0.68            | 0.78            |
|                                                              | PI          | 0.56            | 0.56            | 0.56            | 0.56            | 0.56            | 0.60            | 0.60            | 0.60            | 0.60            | 0.60            | 0.63            | 0.63            | 0.63            | 0.63            | 0.63            | 0.67            | 0.67            | 0.67            | 0.67            | 0.67            |
| -4°F (-20.0°C)                                               | TC          | 16325           | 16346           | 16364           | 16365           | 16387           | 17308           | 17329           | 17346           | 17349           | 17370           | 18342           | 18366           | 18387           | 18390           | 18411           | 19448           | 19469           | 19490           | 19493           | 19517           |
|                                                              | S/T         | 0.62            | 0.73            | 0.83            | 0.83            | 0.94            | 0.59            | 0.69            | 0.78            | 0.78            | 0.89            | 0.55            | 0.65            | 0.74            | 0.74            | 0.84            | 0.52            | 0.61            | 0.70            | 0.70            | 0.79            |
|                                                              | PI          | 0.58            | 0.58            | 0.58            | 0.58            | 0.58            | 0.62            | 0.62            | 0.62            | 0.62            | 0.62            | 0.65            | 0.66            | 0.66            | 0.66            | 0.66            | 0.69            | 0.69            | 0.70            | 0.70            | 0.70            |
| 0°F (-17.8°C)                                                | TC          | 16843           | 16865           | 16884           | 16884           | 16907           | 17858           | 17879           | 17897           | 17900           | 17921           | 18925           | 18949           | 18970           | 18974           | 18995           | 20065           | 20087           | 20109           | 20112           | 20137           |
|                                                              | S/T         | 0.63            | 0.73            | 0.84            | 0.84            | 0.95            | 0.59            | 0.69            | 0.79            | 0.79            | 0.90            | 0.56            | 0.65            | 0.74            | 0.74            | 0.85            | 0.53            | 0.62            | 0.70            | 0.70            | 0.80            |
|                                                              | PI          | 0.59            | 0.59            | 0.59            | 0.59            | 0.59            | 0.63            | 0.63            | 0.63            | 0.63            | 0.63            | 0.66            | 0.67            | 0.67            | 0.67            | 0.67            | 0.70            | 0.71            | 0.71            | 0.71            | 0.71            |
| 5°F (-15.0°C)                                                | TC          | 17491           | 17514           | 17533           | 17534           | 17557           | 18545           | 18567           | 18585           | 18589           | 18611           | 19653           | 19678           | 19700           | 19704           | 19726           | 20837           | 20859           | 20882           | 20885           | 20911           |
|                                                              | S/T         | 0.63            | 0.74            | 0.84            | 0.84            | 0.96            | 0.60            | 0.70            | 0.80            | 0.80            | 0.91            | 0.56            | 0.66            | 0.75            | 0.75            | 0.86            | 0.53            | 0.62            | 0.71            | 0.71            | 0.81            |
|                                                              | PI          | 0.60            | 0.60            | 0.60            | 0.60            | 0.60            | 0.64            | 0.64            | 0.64            | 0.64            | 0.64            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.72            | 0.72            | 0.72            | 0.72            | 0.72            |
| 14°F (-10.0°C)                                               | TC          | 18012           | 18037           | 18056           | 18057           | 18081           | 19098           | 19121           | 19140           | 19143           | 19166           | 20239           | 20265           | 20288           | 20292           | 20314           | 21458           | 21482           | 21505           | 21508           | 21535           |
|                                                              | S/T         | 0.62            | 0.73            | 0.83            | 0.83            | 0.94            | 0.59            | 0.69            | 0.78            | 0.78            | 0.89            | 0.55            | 0.65            | 0.74            | 0.74            | 0.84            | 0.52            | 0.61            | 0.69            | 0.69            | 0.79            |
|                                                              | PI          | 0.53            | 0.54            | 0.54            | 0.54            | 0.54            | 0.57            | 0.57            | 0.57            | 0.57            | 0.57            | 0.60            | 0.60            | 0.60            | 0.60            | 0.60            | 0.64            | 0.64            | 0.64            | 0.64            | 0.64            |
| 17°F (-8.3°C)                                                | TC          | 18186           | 18211           | 18231           | 18231           | 18256           | 19282           | 19305           | 19324           | 19328           | 19351           | 20434           | 20461           | 20484           | 20488           | 20511           | 21666           | 21689           | 21713           | 21716           | 21743           |
|                                                              | S/T         | 0.62            | 0.72            | 0.82            | 0.82            | 0.94            | 0.58            | 0.68            | 0.78            | 0.78            | 0.88            | 0.55            | 0.64            | 0.73            | 0.73            | 0.83            | 0.52            | 0.61            | 0.69            | 0.69            | 0.79            |
|                                                              | PI          | 0.51            | 0.51            | 0.51            | 0.51            | 0.51            | 0.54            | 0.54            | 0.54            | 0.54            | 0.54            | 0.58            | 0.58            | 0.58            | 0.58            | 0.58            | 0.61            | 0.61            | 0.61            | 0.61            | 0.61            |
| 23°F (-5.0°C)                                                | TC          | 18075           | 18099           | 18119           | 18120           | 18144           | 19164           | 19187           | 19206           | 19210           | 19233           | 20309           | 20336           | 20358           | 20362           | 20385           | 21533           | 21557           | 21580           | 21583           | 21610           |
|                                                              | S/T         | 0.61            | 0.72            | 0.81            | 0.81            | 0.93            | 0.58            | 0.68            | 0.77            | 0.77            | 0.88            | 0.54            | 0.64            | 0.72            | 0.72            | 0.83            | 0.51            | 0.60            | 0.68            | 0.68            | 0.78            |
|                                                              | PI          | 0.56            | 0.56            | 0.56            | 0.56            | 0.56            | 0.59            | 0.59            | 0.59            | 0.59            | 0.59            | 0.63            | 0.63            | 0.63            | 0.63            | 0.63            | 0.66            | 0.66            | 0.67            | 0.67            | 0.67            |
| 35°F (1.7°C)                                                 | TC          | 17853           | 17877           | 17896           | 17897           | 17921           | 18928           | 18951           | 18970           | 18973           | 18996           | 20059           | 20085           | 20108           | 20111           | 20134           | 21268           | 21291           | 21314           | 21318           | 21344           |
|                                                              | S/T         | 0.60            | 0.70            | 0.80            | 0.80            | 0.91            | 0.56            | 0.66            | 0.75            | 0.75            | 0.86            | 0.53            | 0.62            | 0.71            | 0.71            | 0.81            | 0.50            | 0.59            | 0.67            | 0.67            | 0.76            |
|                                                              | PI          | 0.65            | 0.65            | 0.65            | 0.65            | 0.65            | 0.69            | 0.69            | 0.69            | 0.69            | 0.69            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.77            | 0.77            | 0.77            | 0.77            | 0.78            |
| 41°F (5.0°C)                                                 | TC          | 17755           | 17779           | 17798           | 17799           | 17823           | 18825           | 18848           | 18866           | 18870           | 18893           | 19950           | 19976           | 19998           | 20002           | 20024           | 21152           | 21175           | 21198           | 21202           | 21228           |
|                                                              | S/T         | 0.61            | 0.72            | 0.82            | 0.82            | 0.93            | 0.58            | 0.68            | 0.77            | 0.77            | 0.88            | 0.54            | 0.64            | 0.73            | 0.73            | 0.83            | 0.51            | 0.60            | 0.69            | 0.69            | 0.78            |
|                                                              | PI          | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.75            | 0.75            | 0.75            | 0.75            | 0.75            | 0.80            | 0.80            | 0.80            | 0.80            | 0.80            | 0.84            | 0.85            | 0.85            | 0.85            | 0.85            |
| 47°F (8.3°C)                                                 | TC          | 17658           | 17682           | 17701           | 17702           | 17726           | 18723           | 18745           | 18763           | 18767           | 18789           | 19841           | 19867           | 19889           | 19893           | 19915           | 21037           | 21059           | 21082           | 21086           | 21112           |
|                                                              | S/T         | 0.63            | 0.73            | 0.83            | 0.83            | 0.95            | 0.59            | 0.69            | 0.79            | 0.79            | 0.90            | 0.56            | 0.65            | 0.74            | 0.74            | 0.85            | 0.53            | 0.62            | 0.70            | 0.70            | 0.80            |
|                                                              | PI          | 0.77            | 0.77            | 0.77            | 0.77            | 0.77            | 0.82            | 0.82            | 0.82            | 0.82            | 0.82            | 0.87            | 0.87            | 0.87            | 0.87            | 0.87            | 0.92            | 0.92            | 0.92            | 0.92            | 0.92            |
| 50°F (10.0°C)                                                | TC          | 17695           | 17718           | 17738           | 17738           | 17762           | 18761           | 18783           | 18802           | 18806           | 18828           | 19882           | 19908           | 19930           | 19934           | 19956           | 21080           | 21103           | 21126           | 21129           | 21155           |
|                                                              | S/T         | 0.62            | 0.73            | 0.82            | 0.82            | 0.94            | 0.58            | 0.68            | 0.78            | 0.78            | 0.89            | 0.55            | 0.65            | 0.73            | 0.73            | 0.84            | 0.52            | 0.61            | 0.69            | 0.69            | 0.79            |
|                                                              | PI          | 0.79            | 0.79            | 0.79            | 0.79            | 0.79            | 0.84            | 0.84            | 0.84            | 0.84            | 0.84            | 0.89            | 0.89            | 0.89            | 0.89            | 0.89            | 0.94            | 0.94            | 0.94            | 0.94            | 0.94            |
| 59°F (15.0°C)                                                | TC          | 17804           | 17828           | 17847           | 17848           | 17872           | 18877           | 18899           | 18918           | 18921           | 18944           | 20004           | 20030           | 20053           | 20056           | 20079           | 21210           | 21233           | 21256           | 21259           | 21286           |
|                                                              | S/T         | 0.60            | 0.70            | 0.80            | 0.80            | 0.91            | 0.56            | 0.66            | 0.75            | 0.75            | 0.86            | 0.53            | 0.62            | 0.71            | 0.71            | 0.81            | 0.50            | 0.59            | 0.67            | 0.67            | 0.76            |
|                                                              | PI          | 0.85            | 0.85            | 0.85            | 0.85            | 0.85            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.95            | 0.95            | 0.95            | 0.95            | 0.95            | 1.01            | 1.01            | 1.01            | 1.01            | 1.01            |
| 68°F (20.0°C)                                                | TC          | 17913           | 17937           | 17956           | 17957           | 17981           | 18992           | 19015           | 19033           | 19037           | 19060           | 20127           | 20153           | 20175           | 20179           | 20202           | 21340           | 21363           | 21386           | 21389           | 21416           |
|                                                              | S/T         | 0.58            | 0.68            | 0.77            | 0.77            | 0.88            | 0.54            | 0.64            | 0.72            | 0.72            | 0.83            | 0.51            | 0.60            | 0.68            | 0.68            | 0.78            | 0.48            | 0.57            | 0.65            | 0.65            | 0.74            |
|                                                              | PI          | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.96            | 0.96            | 0.96            | 0.96            | 0.96            | 1.01            | 1.01            | 1.02            | 1.02            | 1.02            | 1.07            | 1.07            | 1.08            | 1.08            | 1.08            |
| 77°F (25.0°C)                                                | TC          | 18022           | 18046           | 18065           | 18066           | 18091           | 19108           | 19130           | 19149           | 19153           | 19176           | 20249           | 20276           | 20298           | 20302           | 20325           | 21469           | 21493           | 21516           | 21519           | 21546           |
|                                                              | S/T         | 0.55            | 0.65            | 0.74            | 0.74            | 0.84            | 0.52            | 0.61            | 0.70            | 0.70            | 0.80            | 0.49            | 0.58            | 0.66            | 0.66            | 0.75            | 0.47            | 0.55            | 0.62            | 0.62            | 0.71            |
|                                                              | PI          | 0.96            | 0.96            | 0.96            | 0.96            | 0.96            | 1.02            | 1.02            | 1.02            | 1.02            | 1.02            | 1.08            | 1.08            | 1.08            | 1.08            | 1.08            | 1.14            | 1.14            | 1.14            | 1.14            | 1.14            |
| 82°F (27.8°C)                                                | TC          | 17939           | 17963           | 17982           | 17983           | 18007           | 19020           | 19042           | 19061           | 19065           | 19088           | 20156           | 20182           | 20205           | 20209           | 20231           | 21371           | 21394           | 21417           | 21421           | 21447           |
|                                                              | S/T         | 0.56            | 0.66            | 0.75            | 0.75            | 0.85            | 0.53            | 0.62            | 0.70            | 0.70            | 0.80            | 0.50            | 0.58            | 0.66            | 0.66            | 0.76            | 0.47            | 0.55            | 0.63            | 0.63            | 0.71            |
|                                                              | PI          | 1.04            | 1.04            | 1.04            | 1.04            | 1.05            | 1.11            | 1.11            | 1.11            | 1.11            | 1.11            | 1.17            | 1.17            | 1.17            | 1.17            | 1.18            | 1.24            | 1.24            | 1.24            | 1.24            | 1.25            |
| 86°F (30.0°C)                                                | TC          | 17692           | 17715           | 17735           | 17735           | 17759           | 18758           | 18780           | 18799           | 18802           | 18825           | 19878           | 19904           | 19927           | 19930           | 19952           | 21076           | 21099           | 21122           | 21125           | 21152           |
|                                                              | S/T         | 0.58            | 0.69            | 0.78            | 0.78            | 0.89            | 0.55            | 0.65            | 0.74            | 0.74            | 0.84            | 0.52            | 0.61            | 0.69            | 0.69            | 0.79            | 0.49            | 0.58            | 0.65            | 0.65            | 0.75            |
|                                                              | PI          | 1.18            | 1.18            | 1.18            | 1.18            | 1.18            | 1.25            | 1.25            | 1.25            | 1.25            | 1.25            | 1.32            | 1.33            | 1.33            | 1.33            | 1.33            | 1.40            | 1.40            | 1.41            | 1.41            | 1.41            |
| 95°F (35.0°C)                                                | TC          | 17135           | 17158           | 17177           | 17178           | 17201           | 18168           | 18190           |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |

# Climate 5000 Series - Gen 4

18K Btu Single Zone Max Performance System - 4-Way Cassette IDU



# BOSCH

| Heating Performance - Indoor Dry Bulb Temperature at 576 CFM |                                    |                  |                  |                    |                    |                          |                  |                  |                    |                    |
|--------------------------------------------------------------|------------------------------------|------------------|------------------|--------------------|--------------------|--------------------------|------------------|------------------|--------------------|--------------------|
| Outdoor DB(°F)                                               | TC  Total Heating Capacity(BTU/hr) |                  |                  |                    |                    | PI  Power Input(kW)      |                  |                  |                    |                    |
|                                                              | Indoor Conditions(DB °F)           |                  |                  |                    |                    | Indoor Conditions(DB °F) |                  |                  |                    |                    |
|                                                              | 60.8°F<br>(16.0°C)                 | 68°F<br>(20.0°C) | 70°F<br>(21.1°C) | 71.6°F<br>(22.0°C) | 75.2°F<br>(24.0°C) | 60.8°F<br>(16.0°C)       | 68°F<br>(20.0°C) | 70°F<br>(21.1°C) | 71.6°F<br>(22.0°C) | 75.2°F<br>(24.0°C) |
| <b>-22°F(-30.0°C)</b>                                        | 8615                               | 8442             | <b>8400</b>      | 8366               | 8291               | 1.83                     | 1.85             | <b>1.85</b>      | 1.85               | 1.86               |
| <b>-13°F(-25.0°C)</b>                                        | 11178                              | 10955            | <b>10900</b>     | 10856              | 10759              | 1.94                     | 1.96             | <b>1.96</b>      | 1.96               | 1.97               |
| <b>-4°F(-20.0°C)</b>                                         | 15178                              | 14874            | <b>14800</b>     | 14741              | 14608              | 2.24                     | 2.25             | <b>2.26</b>      | 2.26               | 2.27               |
| 0°F(-17.8°C)                                                 | 16181                              | 15857            | 15778            | 15715              | 15573              | 2.42                     | 2.44             | 2.45             | 2.45               | 2.46               |
| <b>5°F(-15.0°C)</b>                                          | 17434                              | 17085            | <b>17000</b>     | 16932              | 16780              | 2.65                     | 2.67             | <b>2.68</b>      | 2.68               | 2.69               |
| 14°F(-10.0°C)                                                | 19895                              | 19497            | 19400            | 19322              | 19148              | 2.63                     | 2.65             | 2.65             | 2.66               | 2.67               |
| <b>17°F(-8.3°C)</b>                                          | 20716                              | 20302            | <b>20200</b>     | 20119              | 19938              | 2.62                     | 2.64             | <b>2.64</b>      | 2.65               | 2.66               |
| 22°F(-5.6°C)                                                 | 20627                              | 20215            | 20114            | 20033              | 19853              | 2.48                     | 2.50             | 2.51             | 2.51               | 2.52               |
| 27°F(-2.8°C)                                                 | 20539                              | 20128            | 20027            | 19947              | 19768              | 2.35                     | 2.37             | 2.37             | 2.38               | 2.39               |
| 32°F(0.0°C)                                                  | 20450                              | 20041            | 19941            | 19861              | 19682              | 2.22                     | 2.24             | 2.24             | 2.24               | 2.25               |
| <b>35°F(1.7°C)</b>                                           | 20397                              | 19989            | <b>19889</b>     | 19809              | 19631              | 2.14                     | 2.15             | <b>2.16</b>      | 2.16               | 2.17               |
| 37°F(2.8°C)                                                  | 20553                              | 20142            | 20041            | 19961              | 19781              | 2.08                     | 2.10             | 2.10             | 2.11               | 2.12               |
| 42°F(5.6°C)                                                  | 20942                              | 20523            | 20420            | 20339              | 20156              | 1.95                     | 1.96             | 1.97             | 1.97               | 1.98               |
| 44.6°F(7.0°C)                                                | 21098                              | 20676            | 20572            | 20490              | 20306              | 1.89                     | 1.91             | 1.91             | 1.92               | 1.92               |
| <b>47°F(8.3°C)</b>                                           | 21331                              | 20905            | <b>20800</b>     | 20717              | 20530              | 1.81                     | 1.83             | <b>1.83</b>      | 1.83               | 1.84               |
| 52°F(11.1°C)                                                 | 20303                              | 19897            | 19798            | 19719              | 19541              | 1.61                     | 1.62             | 1.62             | 1.63               | 1.63               |
| 57°F(13.9°C)                                                 | 19276                              | 18890            | 18796            | 18721              | 18552              | 1.40                     | 1.41             | 1.42             | 1.42               | 1.42               |
| 59°F(15.0°C)                                                 | 18865                              | 18487            | 18395            | 18321              | 18157              | 1.32                     | 1.33             | 1.33             | 1.34               | 1.34               |
| 62°F(16.7°C)                                                 | 18248                              | 17883            | 17794            | 17723              | 17563              | 1.20                     | 1.21             | 1.21             | 1.21               | 1.22               |
| <b>64.4°F(18.0°C)</b>                                        | 17755                              | 17400            | <b>17313</b>     | 17244              | 17088              | 1.10                     | 1.11             | <b>1.11</b>      | 1.11               | 1.12               |
| <b>69.8°F(21.0°C)</b>                                        | 16645                              | 16312            | <b>16231</b>     | 16166              | 16020              | 0.88                     | 0.88             | <b>0.89</b>      | 0.89               | 0.89               |
| <b>75.2°F(24.0°C)</b>                                        | 15535                              | 15224            | <b>15148</b>     | 15088              | 14952              | 0.66                     | 0.66             | <b>0.66</b>      | 0.66               | 0.67               |

Bolded values were tested. All other values are simulated and for reference only.